MeshX platform for Barclays

Technical scope: what runs in your VPC, what is built, and what is not yet started

Purpose of this session

Barclays asked whether Foundation can run in its own VPC, without a meshX-operated control plane

- Leave with a capability map. It shows which services Barclays supplies and which run in the cluster.
- Leave with an honest list of what is built, designed, and not started.
- Agree next steps, with owners, for the parts not yet built.
- Out of scope: commercial terms, staffing, and legal terms.

Capability map

Status of each capability, based on the repository today

Capability	Status	What exists
Stack operator (pull-based reconcile)	**Proven**	Kubernetes operator; Direct (Helm) or Flux mode; dependency-ordered deploys
Helm chart set	**Proven**	23 charts, including CloudNativePG, Redis, Kafka, NATS, Trino and S3-compatible storage
Air-gap bundle import	**Partial**	Upload and import APIs, SHA256 checks, in-cluster registry; image references not yet rewritten
Identity (Logto)	**Partial**	Logto chart and OAuth integration; product templates still default to Auth0
LLM access	**Partial**	LiteLLM gateway; default is the public Anthropic API; Bedrock not yet tested
Cost visibility	**Proven**	Per-request price tables and cost per call; visibility only
Model routing	**Gap**	Host-to-upstream mapping only; no model selection
GPU and on-prem models	**Gap**	No GPU requests; no path to an on-prem model endpoint
Federation across entities	**Gap**	One namespace per tenant; no federation code
Operator access	**Gap**	Only Tailscale found; bastion and break-glass not defined

Release and delivery in an air-gapped VPC

Six steps, with the current state of each

Step	State	Note
1. Build and sign the stack bundle (OCI)	Partial	OCI bundles are supported; signing is not yet verified
2. Push the bundle to the customer registry	Designed	The cluster does not yet pull from a customer registry in an air-gap install
3. Operator detects the new version	Proven	Pull-based reconcile in Direct or Flux mode
4. Customer approval gate	Gap	Not built
5. Promotion to the target stack	Partial	Dependency-ordered apply and pruning; no promotion across environments
6. Rollback to a prior version	Gap	Helm rollback is used only to recover stuck releases

Release cadence: a train agreed with Barclays change management, plus a separate lane for critical security patches. No SLA is quoted yet.

Inference inside the Barclays boundary

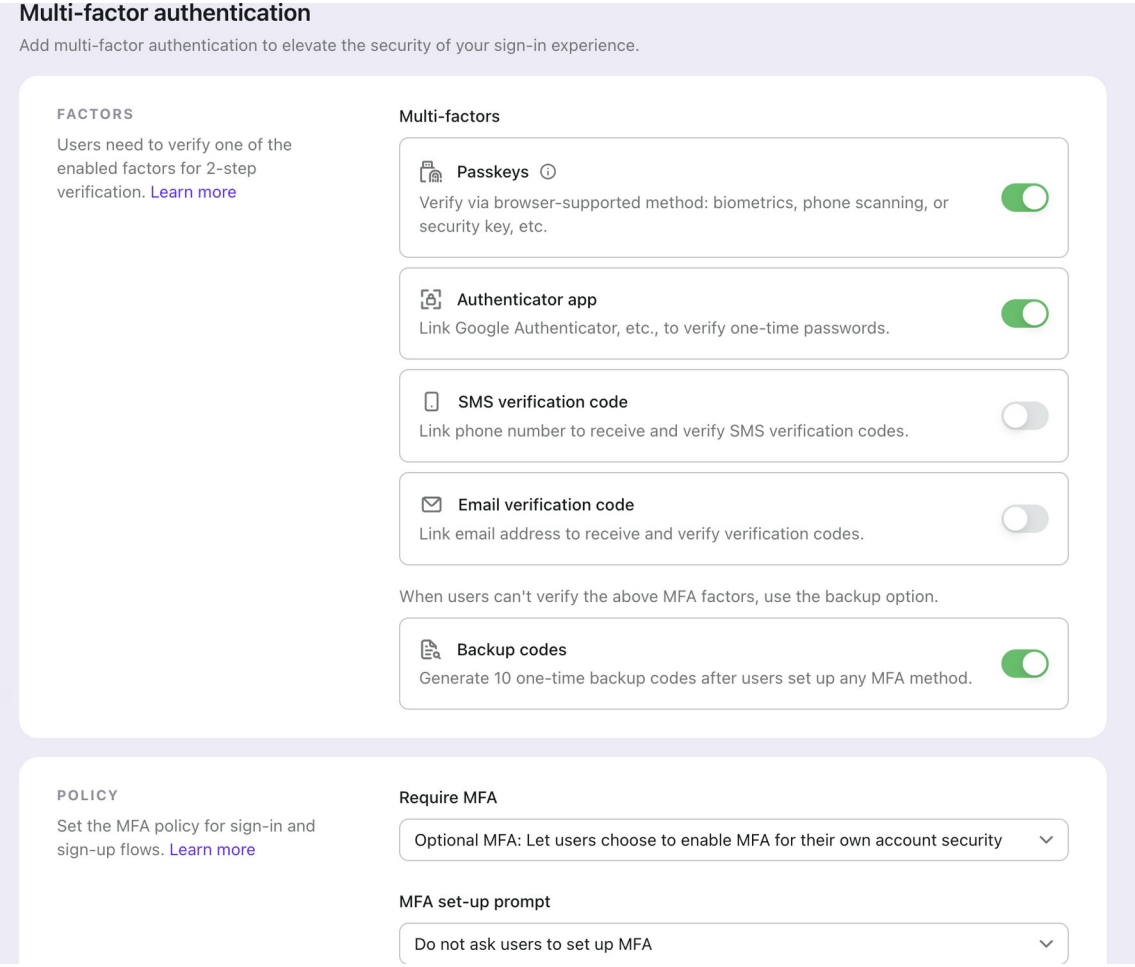
Bedrock-only operation. Not yet verified in a running environment.

Requirement	State	Pass test in the AWS test environment
Bedrock as the LiteLLM provider	**Gap**	A chat request completes with no public endpoint call
Approved model list and region	**Gap**	Only Barclays-approved model IDs are reachable
IAM role for the service account	**Gap**	Requests are signed with the role; no static API key
PrivateLink endpoint and private DNS	**Gap**	The Bedrock name resolves to the private endpoint
Logging and encryption keys	**Gap**	Invocation logs land in the customer account, encrypted with a customer key

The test environment will be built and the results reported before the session.

Identity: sign-in controls

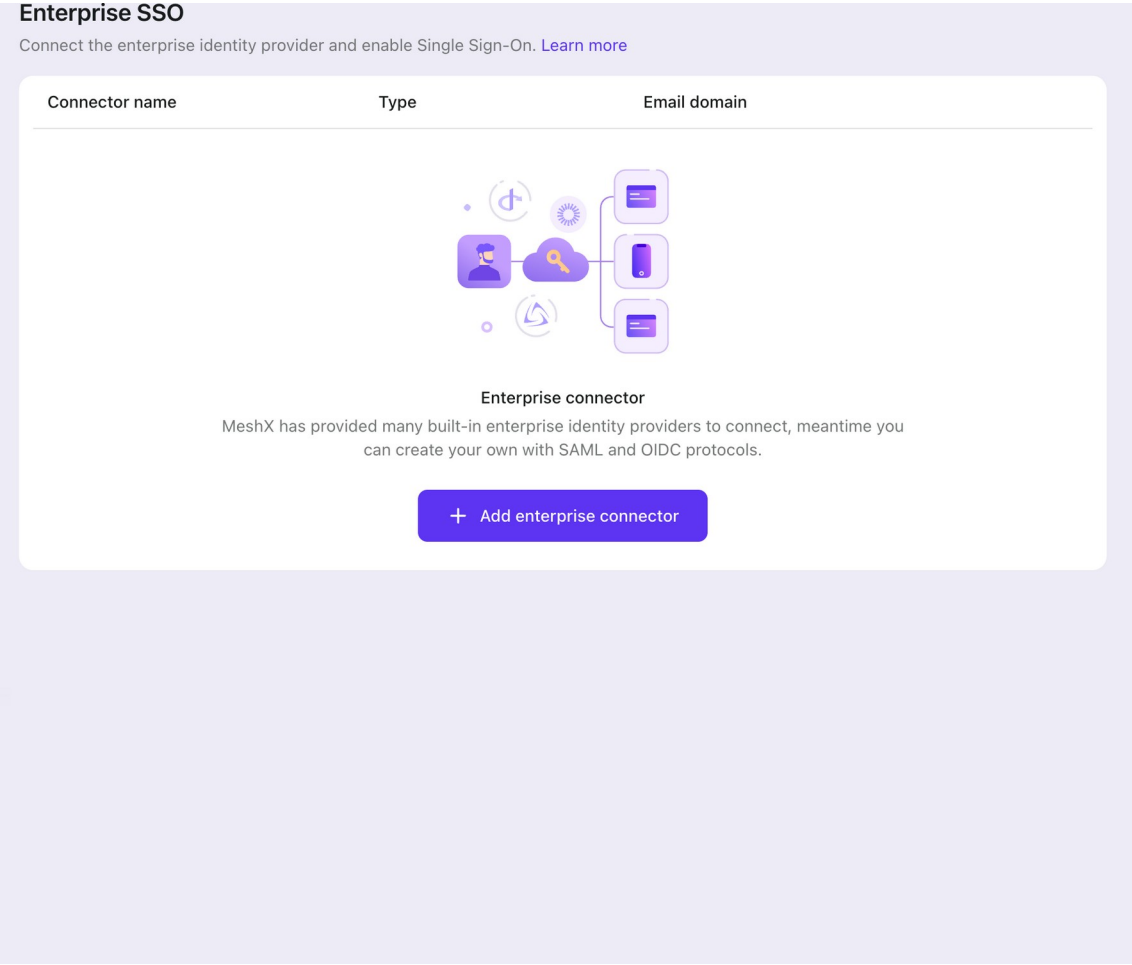
MFA and sign-up controls are set in the admin console



- Set when MFA is required: optional or enforced.
- Select the factors the bank accepts.
- Backup codes give a recovery path. The policy stays strong.
- Auth0 defaults in the product templates must be removed before a Barclays install.

Identity: federation and sign-up control

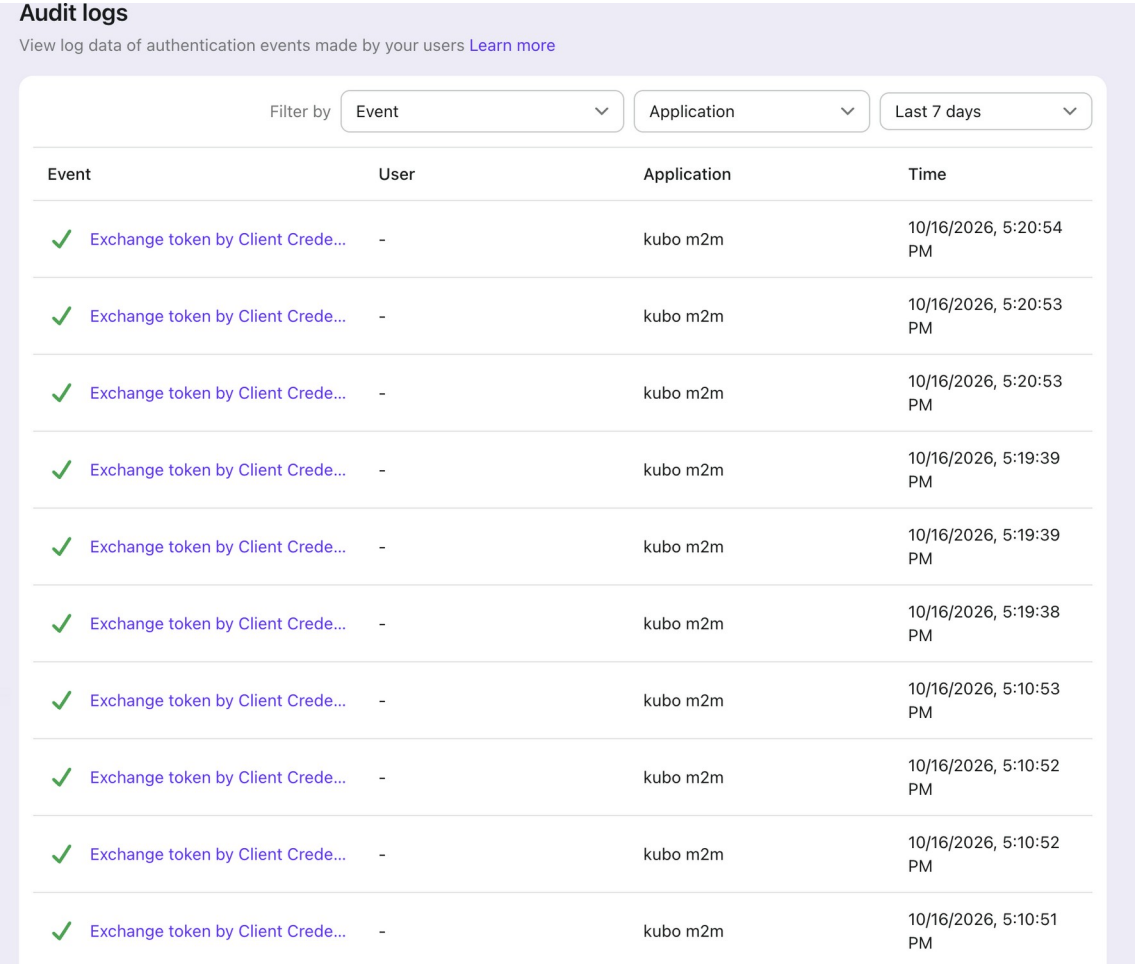
Staff use their identity provider. Sign-up domains are controlled.



- Enterprise SSO connects to SAML and OIDC providers.
- Each connector is linked to an email domain.
- Block disposable and high-risk domains at sign-up.

Audit trail

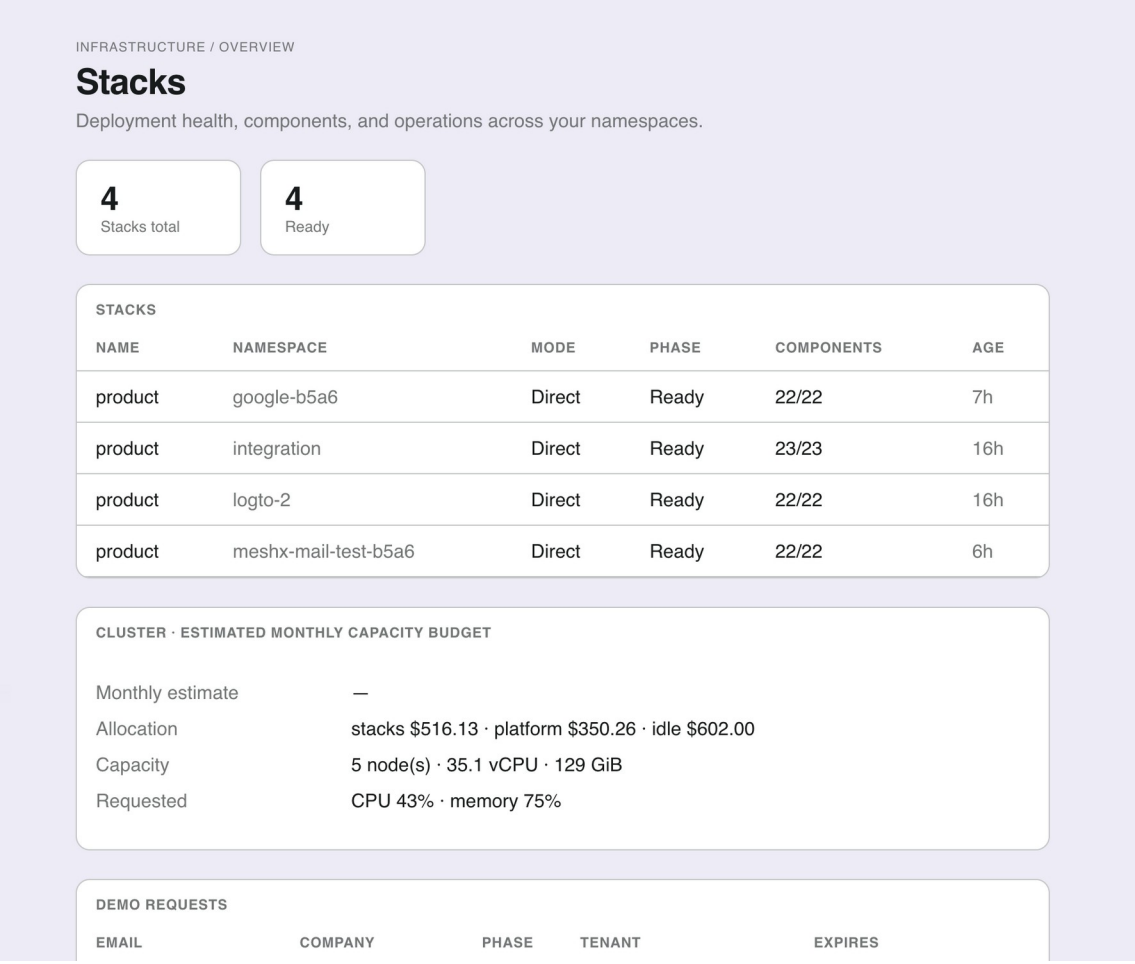
Each token exchange is recorded by application and time



- Filter by event, application, and time.
- Export the log for internal audit and regulator requests.
- Audit evidence inside the cluster needs a mapping to each control. This is to be completed before the review.

Stacks and operations

Stack phase and component status for each environment



- Each stack uses an approved template: Full, Lite, or Dashboard AI only.
- Component status, such as 22/22, shows if a deployment is complete.
- Security hardening: the core charts still need their settings confirmed before we make a hardening claim.

Decisions we need from Barclays

Answers set the order of the work

- Which AWS services are approved, and in which region?
- Does Foundation run in AWS, on-prem, or both?
- Do you need GPU nodes for Foundation, or access to your existing model endpoint?
- Which Bedrock models are approved, and is access by PrivateLink or an internal gateway?
- Who runs the platform day to day: Barclays, meshX, or shared?
- Which operator access model do you require: bastion, break-glass, or neither?
- Is identity federation to your provider needed from day one?
- Where does the Databricks target run, and who owns the network path?

Next steps

Items not yet built, with owners and dates to confirm

Action	Owner	Timing
Prove Bedrock-only operation in a new AWS test environment	meshX	Before the session
Wire the air-gap image pull path to the in-cluster registry, with a no-egress test	meshX platform	To confirm
Default to Logto for air-gap profiles	meshX platform	To confirm
Add a customer approval gate and promotion workflow to the release flow	To assign	To confirm
Build a user-facing rollback workflow	To assign	To confirm
Confirm GPU and on-prem model design after question 3	To assign	After Barclays answers
Choose the Open Sharing network option with Barclays network architects	meshX with Barclays network team	After Barclays answers